

Assignment 1: Machine Learning Based RF (Radio Frequency) Anomaly Detection for Secure Vehicular Networks

The goal of this assignment is to apply machine learning techniques to detect anomalies in vehicular networks by classifying benign and malicious RF signals. Use the provided datasets and perform EDA. The next assignment We will focus on implementing the Random Forest algorithm for classification.

Download the provided datasets 2.4ghz_passivescan_background_loc1.csv (benign) and 3dbm_gaussiannoise_1.csv (malicious) from the Canvas. Follow the steps outlined in the questions below to complete the assignment.

Dataset Overview-

- 2.4ghz_passivescan_background_loc1.csv (benign)- Data was obtained from real-world environments. Represents benign RF signals in normal vehicular network conditions. Captured under realistic conditions with no malicious interference.
- Malicious Dataset: Contains RF signals affected by jamming attacks. Jamming was simulated using Gaussian noise to mimic real-world interference.

1. Dataset Preparation

1.1 Load the provided datasets into two separate DataFrames with the help of pandas

```
In [1]: import pandas as pd
df1 = pd.read_csv("2.4ghz_passivescan_background_loc1.csv")
df2=pd.read_csv("3dbm_gaussiannoise_1.csv")
```

1.2 Combine them into a single DataFrame, adding a new column label where:

0 represents benign signals.

1 represents malicious signals.

```
In [3]: df1["label"] = 0
df2["label"] = 1
ccombined_df = pd.concat([df1, df2], ignore_index=True)
print(combined_df)
```

	freq1	noise	max_magnitude	total_gain_db	base_pwr_db	rsssi	\
0	2412	-108	95	80	387	20	
1	2412	-108	67	76	391	21	
2	2412	-108	79	77	390	21	
3	2412	-108	73	79	387	21	
4	2412	-108	85	78	388	19	
...	...	...	...	...	...	...	
4410	2462	-102	46	74	387	5	
4411	2462	-102	51	74	387	5	
4412	2462	-102	34	74	387	6	
4413	2462	-102	38	74	387	4	
4414	2462	-102	37	74	387	5	

	relpwr_db	avgpwr_db	label
0	13	59	0
1	2	54	0
2	7	55	0
3	17	58	0
4	16	57	0
...	...	...	...
4410	10	49	1
4411	16	50	1
4412	14	48	1
4413	13	49	1
4414	14	48	1

[4415 rows x 9 columns]

```
features = ['freq1', 'noise', 'max_magnitude', 'total_gain_db', 'base_pwr_db', 'rsssi',
            'relpwr_db', 'avgpwr_db']
```

1.3 Create a new column SNR (Signal-to-Noise Ratio), calculated as- (SNR = ['max_magnitude']/['noise']) and append it to features

```
In [6]: combined_df["SNR"] = combined_df["max_magnitude"] / combined_df["noise"]
        print(combined_df["SNR"])
```

```
0    -0.879630
1    -0.620370
2    -0.731481
3    -0.675926
4    -0.787037
...
4410  -0.450980
4411  -0.500000
4412  -0.333333
4413  -0.372549
4414  -0.362745
```

Name: SNR, Length: 4415, dtype: float64

1.4 Display your dataset, check if the 'label' and 'SNR' columns have been added correctly to your dataset

```
In [8]: print("\nColumn Info:")
        print(combined_df.dtypes)
        print("\nSNR Column Statistics:")
        print(combined_df["SNR"].describe())
```

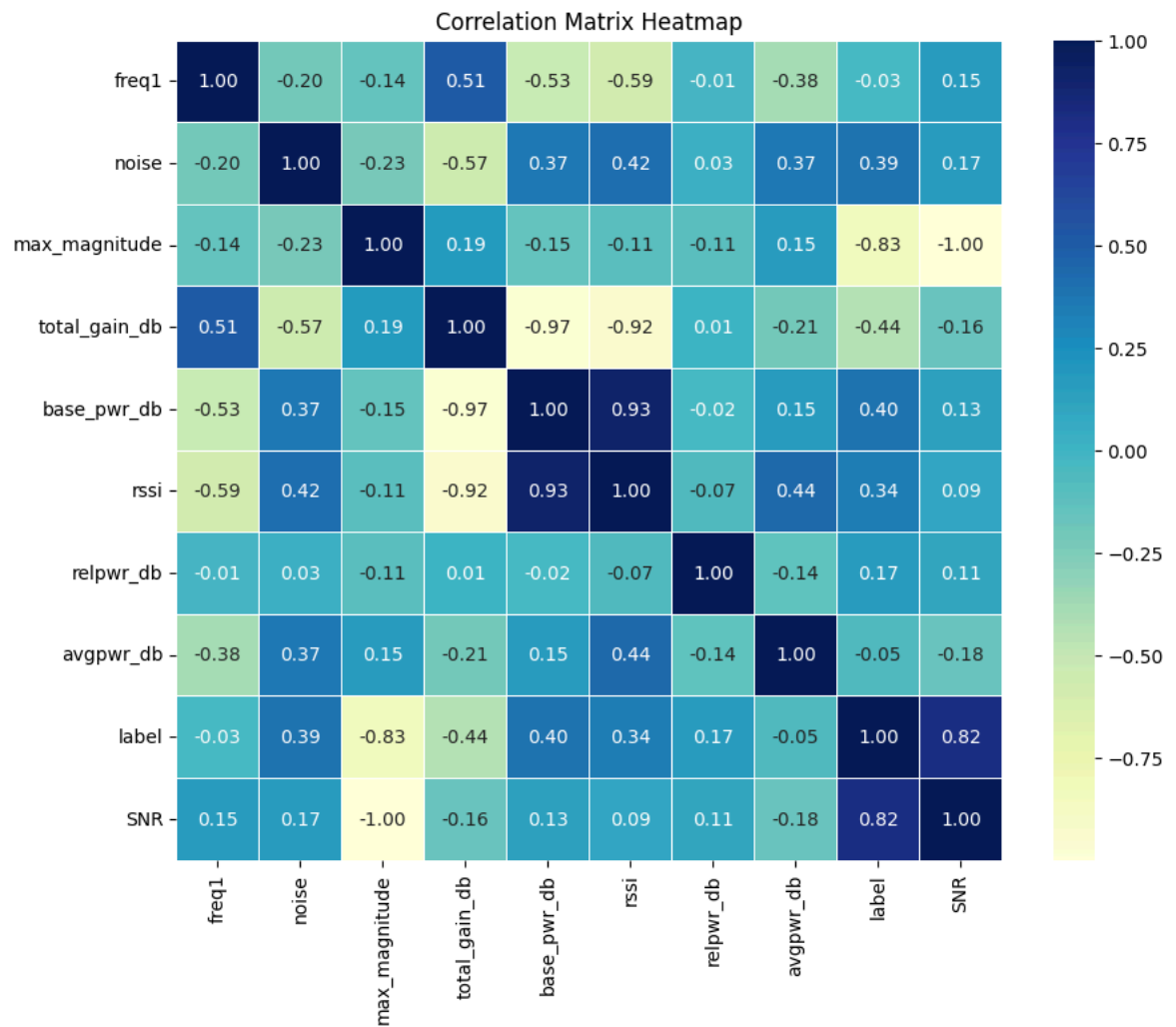
```
Column Info:
freq1          int64
noise          int64
max_magnitude int64
total_gain_db  int64
base_pwr_db    int64
rssi           int64
relpwr_db      int64
avgpwr_db      int64
label          int64
SNR            float64
dtype: object
```

```
SNR Column Statistics:
count    4415.000000
mean     -0.679127
std       0.279618
min      -2.108911
25%      -0.940594
50%      -0.673267
75%      -0.411765
max       -0.155340
Name: SNR, dtype: float64
```

2. Exploratory Data Analysis (EDA)

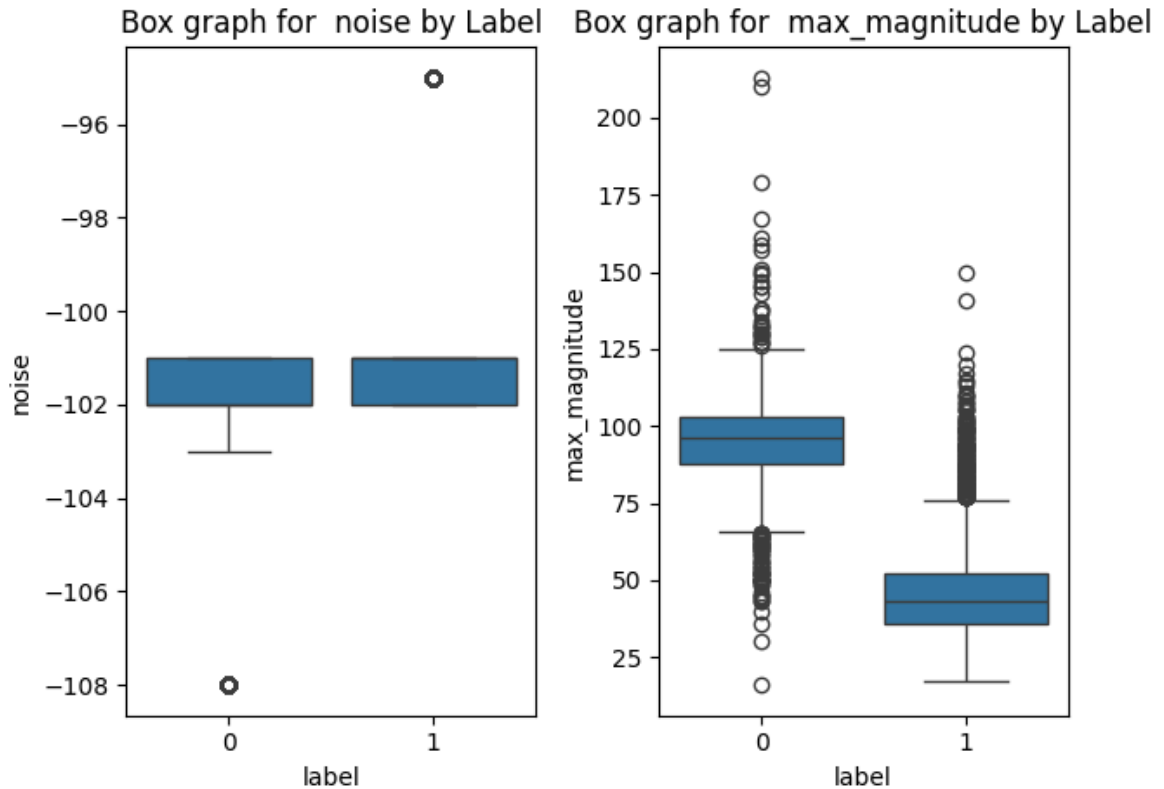
2.1 A correlation matrix heatmap for the features and the target label (label)

```
In [13]: import seaborn as sns
import matplotlib.pyplot as plt
correlation_matrix = combined_df.corr()
plt.figure(figsize=(10, 8))
sns.heatmap(correlation_matrix, annot=True, cmap="YlGnBu", fmt=".2f", linewidths=1)
plt.title("Heatmap of Correlation Matrix")
plt.show()
```



2.2 Box plots for at least two features (noise, max_magnitude, rssi) grouped by label.

```
In [18]: features = ["noise", "max_magnitude", "rssi"]
for i, feature in enumerate(features[:2], 1):
    plt.subplot(1, 2, i)
    sns.boxplot(x="label", y=feature, data=combined_df)
    plt.title(f"Box graph for {feature} by Label")
plt.tight_layout()
plt.show()
```



In []:

2.3 List the key features that distinguish between benign and malicious RF signals based on your EDA insights of correlation matrix

From our exploratory data analysis (EDA), we found some clear differences between benign and malicious RF signals. Malicious signals tend to have higher noise levels, which makes them more distorted and harder to interpret. They also often show higher maximum magnitude values, meaning they can be stronger but also more unstable. Another key factor is the Signal-to-Noise Ratio (SNR)—benign signals usually have a higher SNR, making them clearer and more stable, while malicious signals tend to have lower SNR due to excessive noise. Lastly, the Received Signal Strength Indicator (RSSI) can help differentiate between the two, as malicious signals often show irregular or fluctuating patterns.

2.4 Explain what you understand from Box plots. How features differ in benign and malicious.

The box plots illustrate the differences between two attributes for benign (0) and malicious (1) labels: noise and maximum magnitude. - Noise Feature: The values for both labels are comparable. Although there are a few anomalies, the change is minimal. -Max Magnitude Feature: Benign (0) exhibits greater variance and a higher median. Malicious (1) has fewer outliers and lower values. Accordingly, "max magnitude" is a more effective way to distinguish between benign and malevolent situations.